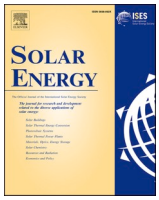
Solar Energy 300 (2025) 113828



Contents lists available at ScienceDirect

Solar Energy

journal homepage: www.elsevier.com/locate/solener





Experimental investigation on improving photovoltaic module efficiency using adaptive fuzzy-PID water cooling

Mohamad Shukor bin Abdul Rahim [a,*], Mohammad Faridun Naim bin Tajuddin [a], Mohd Sazli Saad [b], Md Hasanuzzaman [c], Mohd Zamri Zahir Ahmad [d], Ahmad Ashraf Abdul Halim [d], Ernie Che Mid [a]

[a] Faculty of Electrical Engineering & Technology, Pauh Putra Campus, Universiti Perlis (UniMAP), Pauh, 02600 Arau, Perlis, Malaysia
[b] Faculty of Mechanical Engineering & Technology, Pauh Putra Campus, Universiti Malaysia Perlis (UniMAP), Pauh, 02600 Arau, Perlis, Malaysia
[c] Higher Institution Centre of Excellence (HICoE), UM Power Energy Dedicated Advanced Centre (UMPEDAC), Level 4, Wisma R&D, Universiti Malaya, Jalan Pantai Baharu, 59990 Kuala Lumpur, Malaysia
[d] Faculty of Electronic Engineering & Technology, Pauh Putra Campus, Universiti Malaysia Perlis (UniMAP), Pauh, 02600 Arau, Perlis, Malaysia

ARTICLE INFO

Keywords:
Photovoltaic
Adaptive control
Active water cooling
Real-time feedback
Fuzzy-PID control

ABSTRACT

Photovoltaic (*PV*) modules are important components in solar energy systems, but their efficiency is significantly limited by excessive heat generation due to intense solar irradiance, especially in equatorial climates like Malaysia. This excessive heat not only reduces power conversion efficiency but also accelerates the degradation of *PV* modules, leading to shorter lifespans and higher maintenance costs. This research introduces an adaptive self-tuning Fuzzy-PID controller, optimized using the Sine Cosine Algorithm (SCA), to regulate an active water cooling system for enhanced PV performance. The Sine Cosine Algorithm (SCA) was selected for its efficient exploration–exploitation balance, enabling it to avoid local optima and converge to a global optimum, ensuring better parameter tuning for precise temperature regulation. This cooling system differentiates it from other methods by dynamically adjusting water flow based on real-time temperature feedback. This ensures improved power output while significantly reducing water consumption. Its adaptability surpasses conventional methods, with performance validated against free-flow cooling. The controller-based system effectively reduced the PV module's temperature to an average of 40.0 °C, leading to a 12.2 % increase in power output while conserving water by 60 %. Its adaptive approach ensures optimal cooling efficiency, making it a superior choice for sustainable PV cooling systems.

1. Introduction

Photovoltaic (PV) systems play a crucial role in sustainable energy generation, offering a viable alternative to fossil fuels [1]. However, their efficiency is significantly hindered by excessive heat accumulation, particularly in high-irradiance environments [2]. Elevated temperatures degrade PV module performance, reducing power conversion efficiency and accelerating material wear [3]. To mitigate these effects, effective cooling strategies are essential for maintaining optimal power output and prolonging module lifespan [4]. Conventional cooling methods, including ON-OFF control, manual tuning, and free-flow cooling, lack adaptability and efficient resource management, leading to energy losses and excessive water consumption. This necessitates the development of an intelligent, real-time adaptive cooling mechanism to optimize both thermal regulation and resource utilization. To address the challenges posed by excessive heat, various cooling strategies have been explored to enhance PV efficiency. These approaches can be broadly classified into passive and active cooling methods, each with distinct advantages and limitations. Passive PV cooling utilizes thermodynamic principles without any mechanical equipment or external power source. These systems employ natural means to circulate air or liquid and decrease the system's heat [5]. In contrast, active cooling systems incorporate mechanical and electrical components to effectively reduce heat buildup in PV modules [6].

Active water cooling significantly improves PV system performance by effectively dissipating heat, although it tends to be more complicated than air cooling. Sainthiya [7] optimized water pump energy consumption through interval-based cooling, achieving average

* Corresponding author.
E-mail address: mshukorrahim@unimap.edu.my (M.S. bin Abdul Rahim).

https://doi.org/10.1016/j.solener.2025.113828
Received 28 April 2025; Received in revised form 27 June 2025; Accepted 24 July 2025
Available online 31 July 2025
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temperature reductions that enhanced PV efficiency by around 32.7 % during peak times. Similarly, Lebbi [8] developed a hybrid bi-fluid cooling system that decreased module temperatures by 15 °C and boosted electrical output by 13.17 %, highlighting the benefits of combined cooling and cleaning effects. Hadipour [9] found pulsed-spray cooling to be more resource-efficient, reducing module temperature from 57.1 °C to 24.8 °C with a 33.3 % improvement in power output. Lubon [10] demonstrated that water film cooling increased power output by up to 22 %, achieving temperature drops from 45 °C to 25 °C, while Javidan and Moghadam [11] used jet impingement to lower PV temperatures by 30.3 °C, which resulted in a 47.67 % rise in power output. However, many of these systems operate in open-loop configurations without feedback controls, leading to inefficiencies in water use and suboptimal temperature regulation. For improved efficiency and sustainability, integrating controllers to dynamically adjust water flow based on real-time temperature feedback is crucial, optimizing both cooling performance and resource use.

Classical control method, such as ON-OFF controllers, are commonly used for PV cooling due to their simplicity. This controller activates cooling when temperatures exceed a set threshold, typically around 45 °C, and deactivates when temperatures fall below 35 °C [12]. This approach effectively reduces PV surface temperatures but suffers from inefficiencies due tu temperature and power fluctuation. Various studies have explored specific control strategies to optimize PV cooling. For instance, Yang [13] used a shallow-geothermal system with a U-shaped borehole heat exchanger (UBHE) to recycle cooling water, maintaining module temperatures within an ideal range and reducing energy consumption. Laseinde and Ramere [14] implemented an automatic water spray system with microcontroller-based temperature detection, achieving a 16.65 % increase in PV efficiency by maintaining surface temperatures between 31 °C and 40 °C. Similarly, Mawoli [15] designed a temperature-sensitive solenoid valve system that activates cooling based on a set temperature differential, enhancing cooling efficiency by targeting specific temperature variations. While these classical control strategies improve PV module performance, they often lack adaptability and precise resource management. Adaptive control integration can reduce water and energy consumption while providing more stable temperature control.

Modern control techniques offer adaptable, precise solutions for both simple and complex systems, using fuzzy logic and evolutionary algorithms inspired by human problem-solving and natural processes to regulate linear and nonlinear systems effectively. For instance, fuzzy logic is widely applied in PV systems for energy management, as seen in Loloei and Hosseinian's [16] work on a Floating Photovoltaic System (FPVS), where a PSO-optimized FLC significantly reduced operating costs and power supply losses. Similarly, Qeays [17] developed a hybrid photovoltaic thermal system (HPVTS) using fuzzy TOPSIS and Taguchi methods for multi-response optimization, identifying irradiance as the most critical factor affecting system performance. Bassam [18] demonstrated the utility of an ANFIS model for accurate PV array temperature estimation based on environmental data, achieving a 95 % correlation with experimental data. ANFIS has also been successfully employed for trajectory tracking and disturbance rejection [19]. Fuzzy observer-based control effectively enhances MPPT under varying climatic conditions, ensuring stable and efficient PV power generation [20]. Fuzzy-PID controllers have shown enhanced performance in applications like buck-boost converters, where they improve voltage stability, recovery time, and reduce overshoot compared to traditional PID controllers [21]. In solar tracking systems, fuzzy-PID has been used to optimize rise and settling times, resulting in up to 48.3 % performance improvement for PV modules with tracking systems over fixed setups [22]. Overall, the literature suggests that fuzzy and fuzzy-PID controllers are highly effective in closed-loop systems, enabling faster response times and more efficient energy management.

Controller optimization improves system effectiveness and stability by reducing errors and resource use, using techniques from conventional to metaheuristic algorithms. Conventional optimization methods like the Ziegler-Nichols tuning method are widely used in industry for their structured, robust approach to PID tuning, offering a systematic alternative to trial-and-error, even without a full system model. For instance, Kumar and Patra [23] applied the Ziegler-Nichols method to tune a PID controller for a fourth-order plant, achieving improved dynamic, reduced steady-state error, and lower peak overshoot. Similarly, Meshram and Kanojiya [24] compared traditional Ziegler-Nichols tuning with a modified version (CHR) on DC motors, finding that the modified version reduced rise time, overshoot, and settling time. In renewable energy applications, Chua [25] used a Ziegler-Nichols-tuned PID controller to stabilize the output of a DC-DC boost converter for a horizontal-axis wind turbine, demonstrating enhanced voltage stability even with fluctuating wind speeds.. Boghdady [26] noted that conventional methods like Ziegler-Nichols often fall short of producing optimal parameter values for complex processes. Metaheuristic methods, such as the Sine Cosine Algorithm (SCA), Particle Swarm Optimization (PSO), and Genetic Algorithms (GA), offer adaptive strategies inspired by natural processes, making them effective in handling complex and dynamic problems where traditional methods may fall short [27]. In the context of PV systems, metaheuristic optimization techniques have been employed to regulate water flow in cooling systems, as shown by Shahverdian [28], who used NSGA-II for dynamic multi-objective optimization to enhance PV power output and minimize water usage. The SCA, in particular, has proven effective in real-world applications, such as maximum power point tracking (MPP) in PV systems [29]. Furthermore, metaheuristics have demonstrated their utility in adjusting controller parameters. Nayak [30] used a sine–cosine-enhanced whale optimization algorithm to optimize fuzzy-PID controller settings for power system frequency regulation, achieving improved stability and performance compared to conventional methods. Metaheuristic optimization offers a practical alternative to exhaustive search, making it ideal for optimizing fuzzy-PID controllers and complex systems. The SCA, in particular, has proven effective in handling uncertain search spaces, enhancing control system performance in PV applications and beyond.

To better understand the advancements and limitations of existing PV cooling methods, a comparative analysis of various approaches reported in the literature is presented in Table 1, highlighting their cooling strategies, effectiveness, and associated challenges.

From the above literature review, it is evident that efficient cooling systems and advanced control strategies are important for enhancing the electrical performance of PV modules. Thus, this research aims to:

- Develop a self-tuning fuzzy-PID controller optimized by the Sine Cosine Algorithm (SCA) for active water cooling in PV systems.
- Dynamically enhance thermal regulation and optimize water consumption for sustainable energy generation.
- Experimentally validate the controller's effectiveness in improving PV efficiency and reduce water usage.

The novelty of this work lies in the integration of a fuzzy-PID controller with SCA optimization to achieve adaptive, real-time control of water flow based on PV module temperature. This approach enables responsive and efficient cooling, particularly under fluctuating irradiance conditions, offering a more resource-conscious alternative to conventional fixed-flow systems.

2. Materials and method

2.1. Experimental setup

The experimental rig need to be setup first in order to conduct and contrast the behaviour of PV under different temperature. The experimental setup was designed to simultaneously compare the output performance of the PV modules. Two modules were utilized in the study,

Table 1
Cooling control methods for PV system.

No.	Author	Year	Methods of Cooling	Remarks	Category
1	Safitra et al. [31]	2018	ON-OFF control	Struggles with temperature fluctuations.	ON-OFF control
2	Yang et al. [13]	2019	ON-OFF with timer	Lacks adaptability to dynamic conditions.	
3	Laseinde et al. [14]	2021	ON-OFF control	Inconsistent cooling due to temperature variations.	
4	Lubon et al. [32]	2020	Imitation of rain and water film	Open-loop system, inefficient water use.	Absence of control
5	Lebbi et al. [33]	2021	Hybrid PV/T Bi-fluid Cooling System	No water flow controller, inefficient water use.	
6	Gomaa et al. [34]	2020	Grid-tied PV cooling (Water & Fins cooling)	No water flow controller, lacks front surface cleaning.	
7	Hadipour et al. [35]	2021	Pulsed-spray water cooling	Open-loop system, inefficient water use.	
8	Sainthiya et al. [7]	2021	Interval-Based Back Surface Water Cooling	No feedback system, requires manual adjustments.	Manual tuning
9	Javidan et al. [11]	2021	Water-jet impingement cooling (JIC)	Needs manual control.	
10	Shahverdian et al. [36]	2021	Multi Objective Optimization	The optimization on hourly basis.	Adaptive control (hourly optimize)
11	This research	2025	Fuzzy-PID control using SCA	Adaptive cooling on real-time.	Adaptive control (always optimize)

PV1 served as the reference module without a cooling system, and PV2, equipped with an active water spray cooling system, as illustrated in Fig. 1.



Fig. 1. Real experimental photovoltaic (PV) setup.

The experimental setup consisted of two main parts, which is the piping system configuration and the data acquisition configuration. In the piping system configuration, active water cooling was implemented by spraying water onto the front surface of the PV module using four water ejectors equipped with wide-angle adapters, with the water supplied from a storage tank. Fig. 2 illustrates the water flow route throughout the experiment.

The water spray amount is regulated using a solenoid control valve, which is regulated based on data acquisition activities. The water flow rate is measured by a flowmeter situated beside the control valve. The water sprayed on the PV module is then collected and directed back to the storage tank for cooling. Cooled water is stored in a main tank and reused for spraying onto the PV module. The entire piping system operates in a closed-loop configuration, ensuring continuous water circulation for cooling the PV module.

The activities of the data acquisition system in this research was illustrated in Fig. 3. The computer serves as the central processing unit of the system. It analyzes the received inputs, converts the analog voltage, and generates the necessary outputs for the system. The data acquisition system, NI PCI-6251, collects inputs from temperature sensors installed on photovoltaic module 1 (PV1), photovoltaic module 2 (PV2), and the water flow rate. To ensure an accurate average temperature measurement, three sensors are mounted on the underside of each PV module. The system's output is the operation of the control valve.

The water flow is regulated by adjusting the opening of the control valve. When the temperature difference from the desired temperature is high, the water flow increases, and when the temperature difference is low, the water flow decreases accordingly. The temperature of PV2 will be continuously monitored to ensure that the optimum temperature is maintained, thereby enabling the optimum power generation. This study utilized a polycrystalline silicon PV module.

A flow rate sensitivity analysis was not conducted in this study, as the experiment was performed under naturally fluctuating solar irradiance conditions. These variations cause inconsistent temperature behavior, making it impractical to isolate and compare the effects of fixed water flow rates. Instead, the closed-loop control system was designed to dynamically adjust water flow based on real-time temperature error, allowing more accurate and adaptive thermal regulation.

Maximum rated power for this CS6P-250P photovoltaic (PV) module is 250 W with +5 W (2 %) power tolerance. The specifications of electrical data are shown in Table 2. The temperature plays a significant effect on PV module. Due to the power degradation on the temperature rising, the output power of the PV module will be continuously monitor and recorded on IV tracer to create a proper cooling system for the module. The accuracy and uncertainty of the key measurement parameters are summarized in Table 3, ensuring transparency and reliability of the collected experimental data.

2.2. *Development of self-tuning Fuzzy-PID controller*

Fuzzy logic offers robust handling of uncertainty and variability, making it particularly well-suited for applications such as PV module cooling, where environmental conditions fluctuate. Unlike traditional controllers that rely on precise values, fuzzy logic controllers adapt to varying input degrees, enhancing responsiveness to real-world situations. In this study, a fuzzy-PID controller was implemented using fuzzification, inference, and defuzzification processes. The initial stage, fuzzification, translates crisp inputs into fuzzy data using Membership Functions (MFs). The fuzzy inference process combines these inputs with control rules to generate fuzzy outputs. Finally, defuzzification converts these outputs back into crisp values, creating actionable control signals. To optimize the controller for dynamic PV cooling, a self-tuning fuzzy-PID strategy was employed with error (e) and change of error (Δe) as inputs, while the PID gains (kp, ki, kd) acted as outputs as shown in Fig. 4. The PV plant modeling using system identification is a continuation of a previous experiment [37], where system parameters were

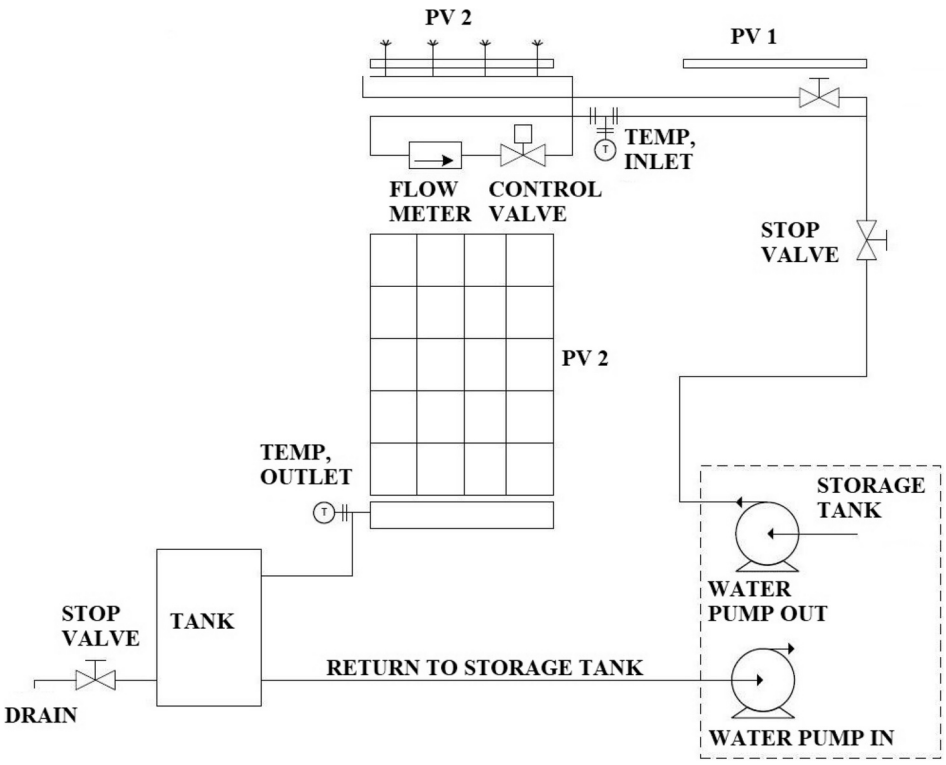


Fig. 2. Piping System Configuration.

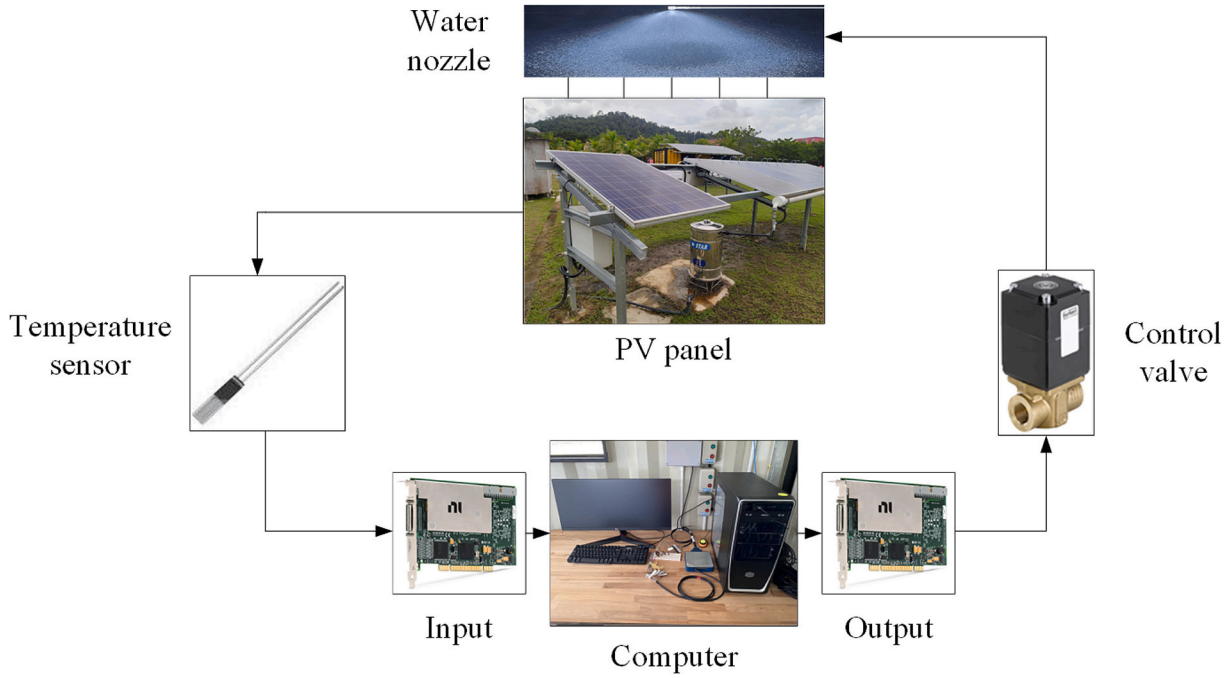


Fig. 3. Data Acquisition System Activities.

identified and validated to ensure accurate temperature response modeling.

These inputs and outputs were mapped into seven triangular MFs, chosen for their rapid responsiveness, and categorized into linguistic terms such as Negative Big (NB), Negative Medium (NM), Negative Small (NS), Zero (ZO), Positive Small (PS), Positive Medium (PM), and Positive Big (PB), as seen in Fig. 5.

The fuzzy rules in Table 4 govern the relationship between the input variables, which is temperature error (e) and its rate of change of error (Δe) and the output PID correction parameters. A total of 147 IF-THEN rules were formulated based on expert knowledge to enable dynamic control under varying thermal conditions. Each rule corresponds to a triplet of fuzzy values representing adjustments for Kp, Ki, and Kd. For instance, when e is Negative Big (NB) and Δe is NB, the system applies the rule PB/NB/PS, which sets Kp to Positive Big (PB), Ki to Negative Big (NB), and Kd to Positive Small (PS). This reflects an aggressive

Table 2
Photovoltaic (PV) module electrical data specifications.

Model	CS6P-250P
Nominal Maximum Power (Pmax)	250 W
Optimum Operating Voltage (Vmp)	30.1 V
Optimum Operating Current (Imp)	8.30 A
Open-circuit voltage (Voc)	37.2 V
Short-circuit current (Isc)	8.87 A
Temperature Coefficient Pmax	−0.43 %/°C
Temperature Coefficient Voc	−0.34 %/°C
Temperature Coefficient Isc	0.065 %/°C
Temperature Coefficient NOCT	45 °C

Table 3
Summary of measurement instruments and estimated uncertainties.

Parameter	Instrument/ Method	Resolution/ Accuracy	Estimated Uncertainty
Temperature (PV Surface)	RTD Sensors	±0.5 °C	±0.5 °C
Water Flow Rate	Flowmeter	±0.1 L/min	±2 %
Voltage (V)	I-V Tracer	±0.5 V	±0.5 V
Current (A)	I-V Tracer	±0.1 A	±0.1 A

proportional response to quickly reduce temperature deviation, a suppressed integral component to prevent overshooting, and a small derivative term for mild damping. The fuzzy inference engine continuously evaluates these inputs and determines the appropriate control action, adjusting the valve opening to regulate water flow. This means that when the temperature error is large, the water flow rate is increased to accelerate cooling, while smaller errors result in lower flow. This real-time modulation allows the system to avoid using the maximum flow rate unnecessarily, contributing to both temperature stability and water conservation. Following fuzzy inference, defuzzification was performed using the Center of Gravity (COG) method. COG, which computes a weighted average of MFs, was selected for its effectiveness in delivering smooth, precise control actions. The COG defuzzification method is described by Equation (1). This defuzzification technique ensures that the fuzzy outputs are transformed back into precise PID control values for real-time adjustments. By employing this fuzzy-PID structure, the controller dynamically adjusts its parameters in response to environmental changes, resulting in enhanced temperature control and efficiency in PV cooling applications.

$$CenterofGravity(COG) = \frac{\sum_x \mu(x) \times x}{\sum_x \mu(x)} \quad (1)$$

Table 4
Fuzzy control rules for PID correction parameters.

Δe / e	NB	NM	NS	ZO	PS	PM	PB
NB	PB/ NB/PS	PB/ NB/PS	PM/ NB/ZO	PM/ NM/ ZO	PS/ NM/ ZO	PS/ ZO/PB	ZO/ ZO/PB
NM	PB/ NB/NS	PB/ NB/NS	PM/ NM/NS	PM/ NM/NS	PS/NS/ ZO	ZO/ ZO/PS	ZO/ ZO/PM
NS	PM/ NM/ NB	PM/ NM/ NB	PM/ NS/NM	PS/NS/ NS	ZO/ ZO/ZO	NS/ PS/PS	NM/ PS/PM
ZO	PM/ NM/ NB	PS/NS/ NM	PS/NS/ NM	ZO/ ZO/NS	NS/PS/ ZO	NM/ PS/PS	NM/ PM/PM
PS	PS/NS/ NB	PS/NS/ NM	ZO/ ZO/NS	NS/PS/ NS	NS/PS/ ZO	NM/ PM/PS	NM/ PM/PS
PM	ZO/ ZO/ NM	ZO/ ZO/NS	NS/PS/ NS	NM/ PM/NS	NM/ PM/ZO	NM/ PB/PS	NB/ PB/PS
PB	ZO/ ZO/PS	NS/ ZO/ZO	NS/PS/ ZO	NM/ PM/ZO	NM/ PB/ZO	NB/ PB/PB	NB/ PB/PB

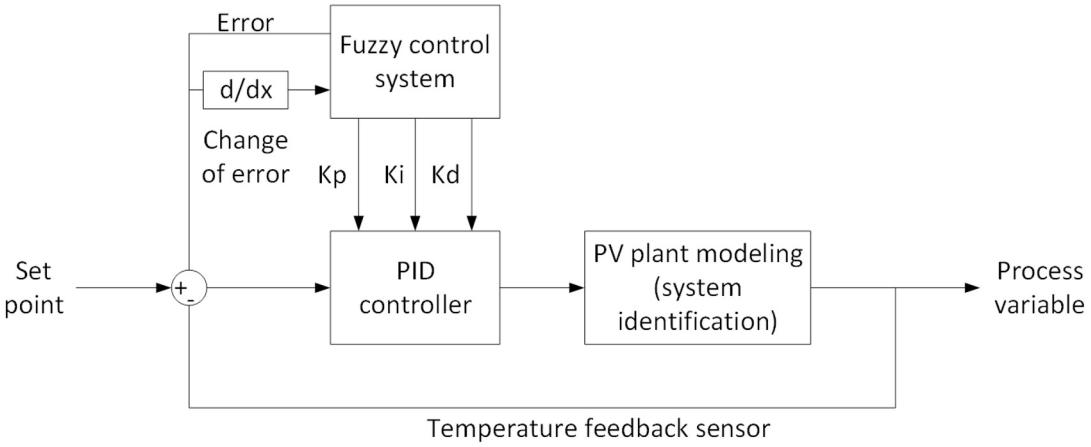


Fig. 4. Fuzzy-PID tuning strategy.

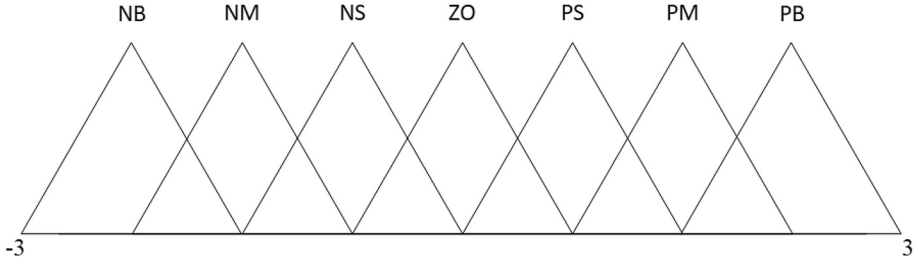


Fig. 5. Membership function.

2.3. Self-tuning fuzzy-PID controller tuning using Sine Cosine Algorithm (SCA)

This research employs a self-tuning Fuzzy-PID controller optimized by the Sine Cosine Algorithm (SCA) to enhance PV module cooling control. The SCA dynamically adjusts the quantization and scaling factors within the fuzzy-PID algorithm, improving control accuracy and robustness. This section presents a fuzzy-PID positioning controller based on the self-tuning SCA. SCA optimizes the quantization and scaling parameters in the fuzzy PID algorithm to maximize actuator accuracy and durability. The SCA is a population-based optimization technique that iteratively updates candidate solutions to converge on a global optimum. The position update mechanism in SCA is governed by the following Eqs. (2) and (3).

$$X(i,j)_{t+1} = X(i,j)_t + r1 \times \sin(r2) \times |r3.\ \{P(j)_t - X(i,j)_t\}| \quad (2)$$

$$X(i,j)_{t+1} = X(i,j)_t + r1 \times \cos(r2) \times |r3.\{P(j)_t - X(i,j)_t\}| \quad (3)$$

where $X(i,j)_t$ represents position of the current solution in *i-th* dimension at *t-th* iteration, $P(j)_t$ represents position of the best individual in *j-th* dimension at *t-th* iteration, | | represent absolute value and the value or r1, r2 and r3 are random numbers.

These two equations are then combined to form Eq. (4). The formula is choose by the random number of r4 which from 0 to 1.

$$X(i,j)_{t+1} = \begin{cases} X(i,j)_t + r1 \times \sin(r2) \times |r3.\ \{P(j)_t - X(i,j)_t\}|; if r4 < 0.5 \\ X(i,j)_t + r1 \times \cos(r2) \times |r3.\ \{P(j)_t - X(i,j)_t\}|; if r4 > 0.5 \end{cases} \quad (4)$$

Equation selection depends on a random variable *r4*; if $r4 < 0.5$, the sine function is used, otherwise, the cosine function is applied. Adaptability is maintained by varying *r1* throughout the iterations using Eq. (5), gradually shifting from exploration to exploitation. The process flow of the algorithm is shown in Fig. 6.

$$r1 = a - a \times \frac{iter}{itermax} \quad (5)$$

The fuzzy inference module within the Fuzzy-PID controller is designed to dynamically regulate PID scaling coefficients based on real-time error (e) and error rate (Δe) inputs, as shown in Fig. 7. To enable adaptive control, the fuzzy membership function scaling factors are optimized using the Sine Cosine Algorithm (SCA), which adjusts the controller response to maintain temperature within desired thresholds and enhance system efficiency. The SCA was configured with a population size of 20 and 100 iterations, optimizing five parameters, two input quantization factors and three output scaling parameters. As a stochastic method, SCA relies on random initialization; therefore, multiple runs were conducted to ensure generalizable and robust optimization. The fitness function minimized temperature error to achieve effective thermal regulation in the PV cooling process.

The Simulink model shown in Fig. 8 illustrates the system's closed-loop operation, where the adaptive fuzzy-PID controller continuously regulates the cooling process based on real-time sensor feedback. The controller is designed to minimize error as quickly as possible, accelerating convergence toward the desired temperature setpoint and improving overall system responsiveness. By utilizing SCA optimization, the fuzzy-PID controller effectively adapts to changing environmental conditions, ensuring sustainable and optimal PV cooling performance.

3. Results and discussion

3.1. Reference temperature selection

The experiment started with selecting a reference temperature, systematically testing various desired settings to determine the optimal parameter for the closed-loop control system of PV modules. This helped identify the most effective temperature to balance better power output and lower water consumption for active cooling control. Each temperature was maintained for 20 min to thoroughly assess its impact on system performance and water consumption. The gathered experimental data, thoroughly detailed in Table 5, serves to evaluate the balance between maximizing power output and minimizing water consumption. At 70 °C, without cooling, no power boost occurred, serving as a baseline. Cooling at 60 °C and below improved performance, with a 50 °C set point yielding a 13.33 W power boost and 19.41 L of water consumption. At 45 °C, the system achieved 15.99 W with 35.66 L of water, while 40 °C provided 17.78 W with higher water use but balanced efficiency gains. Although 35 °C delivered the highest power boost (19.89 W), excessive water usage due to continuous valve operation made it unsustainable. Thus, 40 °C was identified as the optimal temperature for effective and sustainable cooling.

To further justify the improvement in PV module efficiency, a 60-minute experimental analysis was conducted comparing modules with and without controller-based cooling. As shown in Fig. 9, the module operating without cooling averaged around 10.5 % efficiency, with

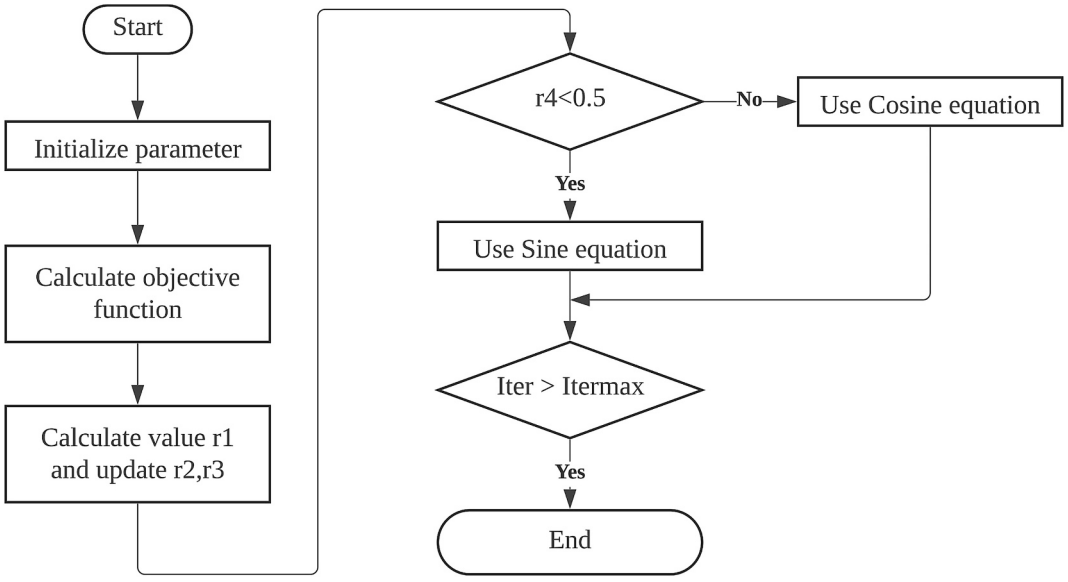


Fig. 6. SCA flowchart.

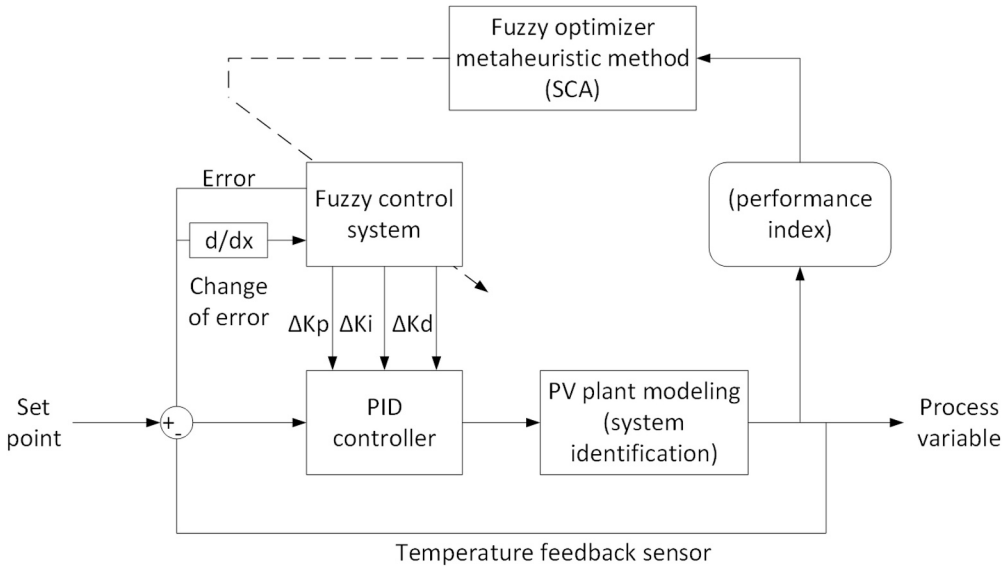


Fig. 7. Fuzzy-PID using SCA tuning strategy.

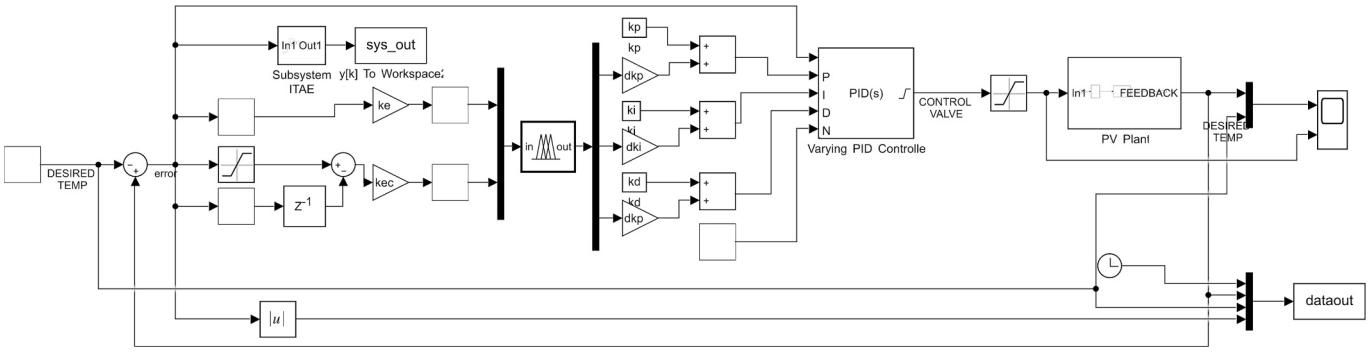


Fig. 8. Fuzzy-PID simulink block.

Table 5
Optimum reference temperature test.

Desired temperature (°C)	Power with Cooling (W)	Power without Cooling (W)	Power Boost (W)	Water Consumption (L)
70	157.83	161.61	0	0
60	168.48	164.32	4.16	6.07
50	177.98	164.65	13.33	19.41
45	181.04	165.05	15.99	35.66
40	173.34	155.56	17.78	68.82
35	185.26	165.37	19.89	Continuous

noticeable drops due to rising temperatures. In contrast, the module with controller-based cooling achieved a higher and more stable average efficiency of approximately 11.76 %. This demonstrates the positive impact of real-time thermal regulation on PV conversion performance.

3.2. Short-term performance: one-hour experimental test

This section evaluates the performance of PV modules under three distinct thermal management conditions, without cooling, with SCA-tuned Fuzzy-PID controller-based cooling, and with free-flow cooling. The analysis focuses on changes in temperature and electrical output before and after the application of these cooling strategies to determine the controller's effectiveness in optimizing PV module performance.

Fig. 10 provide a critical analysis of the thermal dynamics of PV modules under the influence of two cooling strategies. The module denoted as PV1, illustrated by the red line, operates without cooling intervention. In contrast, PV2, indicated by the blue line, benefits from the application of a fuzzy-PID controller-based cooling system before transitioning to a free-flow cooling approach. These strategies are assessed against the environmental conditions, where ambient temperature and solar irradiance are measured to provide a context for the thermal behavior of the PV modules. The solar irradiance remained remarkably stable throughout the experiment, exhibiting minimal fluctuations within the range of 950–1000 W/m^2. This consistency in solar energy input reinforces the validity of the comparative analysis between the cooling strategies, ensuring that the observed thermal effects on the modules are due to the implemented cooling methods rather than environmental variability. PV1 recorded temperatures that underscore the critical need for cooling strategies. It reached a minimum of 57.35 °C, an average that soared to 61.10 °C, and experienced peaks at 64.11 °C. These high temperatures clearly illustrate the risks associated with uncooled modules which not only they are prone to significant efficiency drops, but they are also at risk of suffering faster wear and tear of the material components.

At Time T1, the fuzzy-PID controller-based cooling system was activated with a target setpoint of 40 °C. The controller demonstrated impressive precision in regulating the module temperature, achieving a minimum of 39.28 °C, an average of 39.95 °C, and a peak of 40.60 °C. These results indicate a tightly controlled range of ± 0.60 °C around the target setpoint, with minimal undershoot and overshoot. Such precise

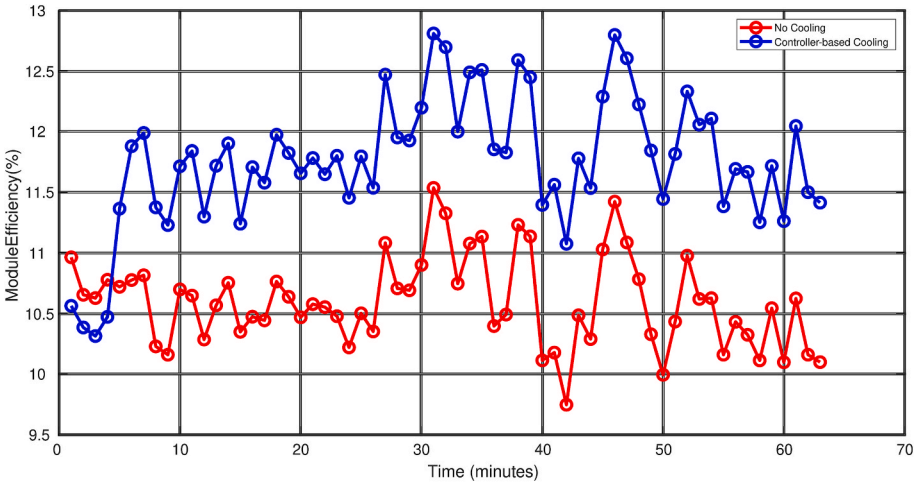


Fig. 9. PV Module efficiency for controller-based cooling and no cooling.

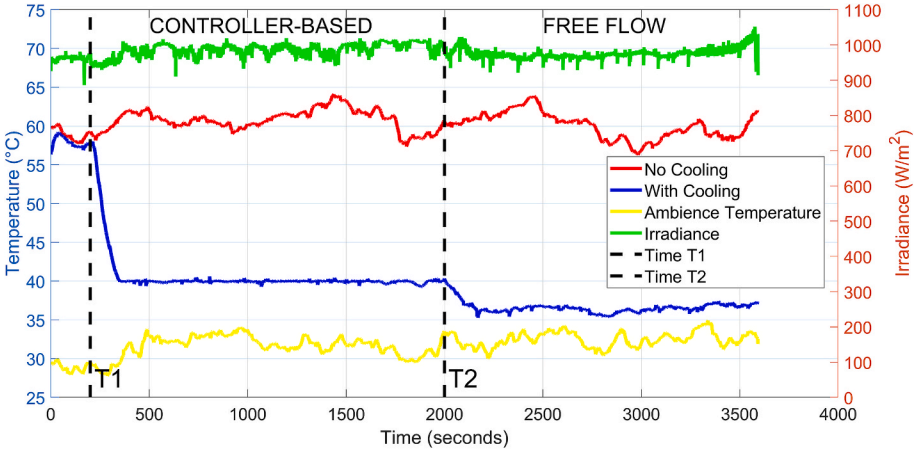


Fig. 10. Temperature comparison of PV panels and irradiance.

control, achieved under outdoor conditions with variable environmental factors, underscores the robustness and reliability of the SCA-tuned fuzzy-PID cooling approach. At Time T2, the cooling strategy transitioned to free-flow cooling, characterized by fully open control valves. This unregulated approach produced a broader temperature range, with a minimum of 35.25 °C, an average of 36.56 °C, and a maximum of 40.26 °C. The significant temperature dip observed with free-flow cooling demonstrates its capability for enhanced thermal reduction. The free-flow approach achieved only marginally better cooling than the controller-based method. Controller-based cooling maintains target temperatures efficiently while conserving water, unlike free-flow cooling, which consumes significantly more water.

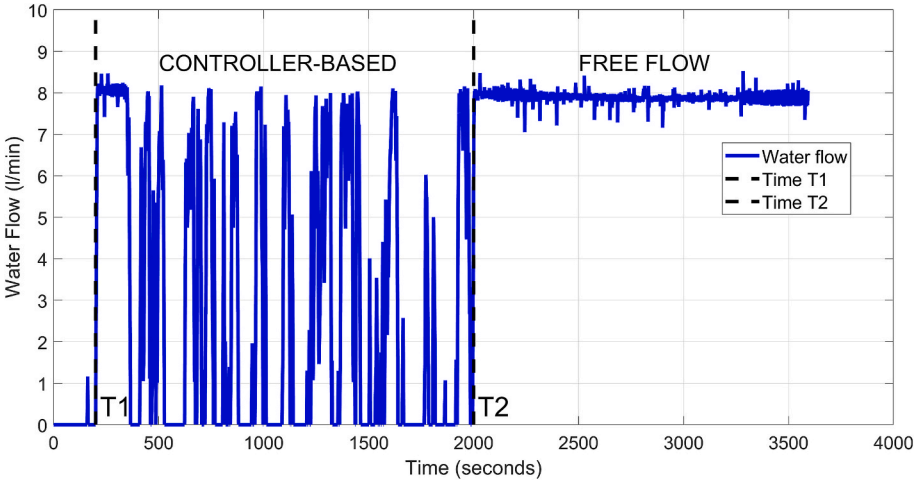


Fig. 11. Water flow rates for controller-based and free flow cooling.

Fig. 11 provides a comparative analysis of water usage between the two cooling strategies. After Time T1, the controller-based system employed an adaptive water flow strategy, consuming 78.5 L of water to maintain the module's target temperature. This method demonstrated high efficiency by using water only when necessary, minimizing wastage. It is important to note that the 8 L/min value refers to the peak flow rate and was only recorded during brief high-temperature spikes. The flow was not sustained, as the controller continuously adjusted valve opening based on real-time temperature feedback. Following the transition to free-flow cooling at Time T2, water consumption surged to 197.6 L over the same duration. This represents a 60.26 % increase in water usage compared to the controller-based system. Additionally, the system operates in a closed-loop configuration, where sprayed water is collected and reused, further enhancing water use efficiency and system sustainability. While free-flow cooling achieved marginally greater thermal reduction, its excessive water consumption highlights a critical limitation in scenarios where water resources are constrained.

Fig. 12 highlights the voltage stability of PV modules under three thermal management strategies. Without cooling, the module exhibited significant instability (32.46–32.95 V), emphasizing the need for thermal regulation. At Time T1, the fuzzy-PID controller-based cooling improved stability significantly (35.05–35.52 V) by maintaining precise temperature control. At Time T2, free-flow cooling slightly increased voltages further (35.37–36.02 V), but the marginal gains were offset by excessive water consumption.

Fig. 13 compares PV module power outputs under different cooling strategies. Without cooling, power ranged from 160.74 W to 166.81 W, averaging 164.30 W, highlighting thermal inefficiency. At Time T1, fuzzy-PID controller-based cooling improved output to 180.33–188.47 W, averaging 184.72 W, demonstrating stable performance. At Time T2, free-flow cooling slightly increased power to 186.84–194.61 W, averaging 190.40 W, a 3.08 % gain over the controller-based method. However, the marginal improvement does not justify the significantly higher water consumption. This trend demonstrates a direct correlation between the stabilized voltage and increased power output. The improved voltage regulation achieved through active cooling contributed significantly to the higher power levels observed under controller-based and free-flow strategies.

Table 6 highlights PV module performance under varying cooling conditions. Without cooling, the module averaged 164.3 W at 61.1 °C, generating 66.96 Wh over 25 min. Controller-based cooling reduced the temperature to 40.0 °C, boosting power by 12.2 % to 184.7 W with an energy output of 75.13 Wh, using just 78.5 L of water. Free-flow cooling lowered the temperature further to 36.6 °C, increasing power by 16.2 % to 190.4 W and generating 77.83 Wh but required 197.6 L of water. While free-flow cooling achieved slightly higher output, its excessive water usage makes the controller-based system a more efficient and sustainable solution.

The analysis of the trade-offs between cooling effect, water utilization, and power enhancement identifies the controller-based cooling as the superior choice. It ensures an elevated level of power efficiency with a sustainable approach to water usage. Voltage stability across the methods further strengthens the case for active cooling, as even modest voltage gains can aggregate to significant improvements in power stability.

3.3. Long-term performance: whole-day experimental test

This subchapter presents the results of whole-day experiments conducted over three days to analyze the impact of solar irradiance fluctuations on PV modules and the real-time response of the cooling controller. Conducting tests on different days allows for evaluating performance under varying weather conditions, including clear skies and intermittent clouds. A full-day experiment was conducted on March 14, 2024, under semi-cloudy conditions to assess the SCA-Fuzzy-PID cooling system's adaptability. Fig. 14 shows the temperature variations of PV modules with and without cooling, alongside solar irradiance changes. The controller was activated between T1 and T2, maintaining the PV module at 40 °C, while the non-cooled module exceeded 65 °C. When irradiance dropped in the late afternoon, the system automatically deactivated cooling, optimizing water use. Condition C1, represents condition where the PV module temperature dropped below the 40 °C threshold due to fluctuations in solar irradiance. Since the cooling system is designed to regulate excessive heat, it deactivates automatically when temperatures fall below the setpoint.

Fig. 15 illustrates the water flow pattern throughout the experiment. The system remained inactive before T1 and after T2, demonstrating efficient water management. During the active phase, water flow adjusted dynamically, peaking at 8 L per minute in response to rising temperatures. This adaptive control mechanism ensured temperature stability while minimizing water consumption.

The experimental data from March 18, 2024 shown in Fig. 16 highlights the temperature regulation of PV modules under cloudy day irradiance conditions. The uncooled module reached a peak temperature of 63 °C, whereas the controller-based cooling maintained an average of 40 °C, effectively mitigating overheating during high irradiance periods. During the inactive phases before T1 and after T2, cooling was unnecessary as temperatures remained below the setpoint. When activated, the controller efficiently stabilized PV module temperature despite irradiance fluctuations. However, periods B1 and B2 (marked as Box 1 and Box 2) reveal conditions where irradiance dips caused temperatures to drop below 40 °C, leading to temporary cooling deactivation. Since

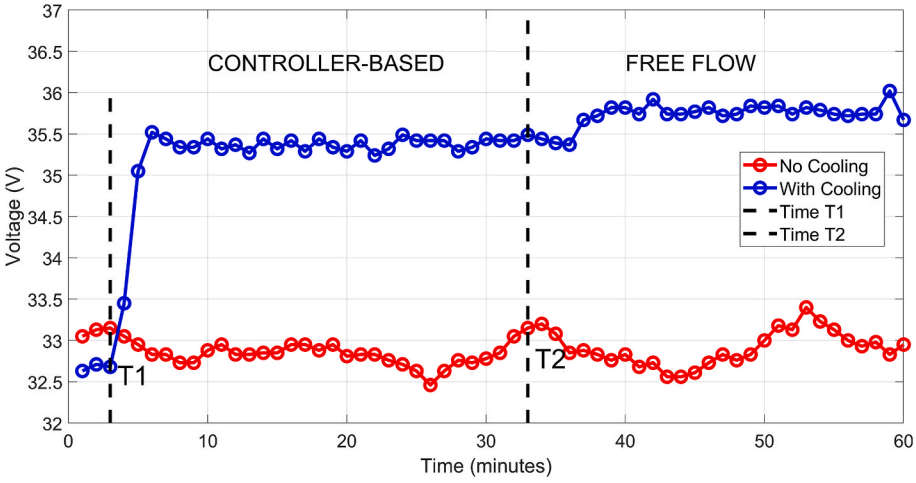


Fig. 12. Open circuit voltage for controller-based cooling and free flow cooling.

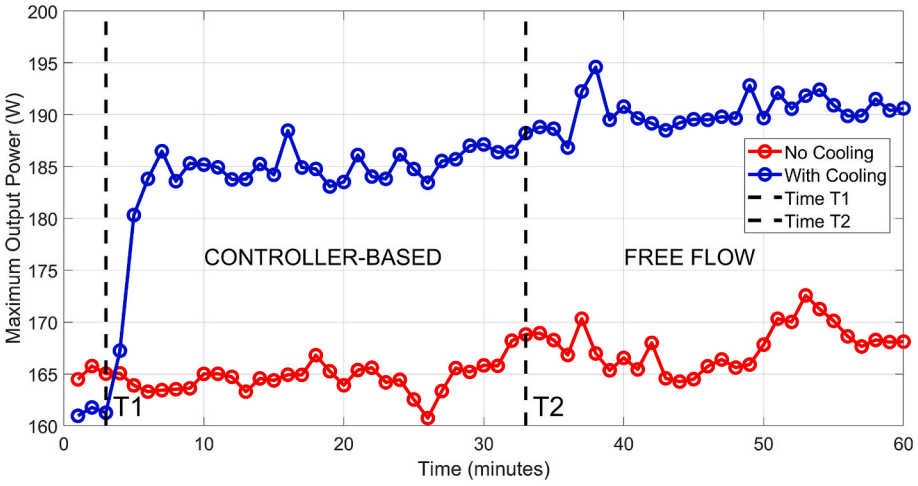


Fig. 13. Maximum output power generated for controller-based cooling and free flow cooling.

Table 6
Power improvement of different cooling techniques.

	Temperature (°C)		Power (W)			Voltage (V)			Water volume (liter)	Power increase (%)			Total energy generated (W. hour)
	Avg	Max	Min	Avg	Max	Min	Avg	Max		Min	Avg	Max	
Without cooling	61.1	64.1	160.7	164.3	166.8	32.5	32.8	33.0	–	0.00	0.00	0.00	66.96
Controller-based cooling	40.0	40.6	180.3	184.7	188.5	35.0	35.4	35.5	78.5	12.2	12.4	13.0	75.13
Free-flow cooling	36.6	40.3	186.8	190.4	194.6	35.4	35.8	36	197.6	16.2	15.9	14.1	77.83

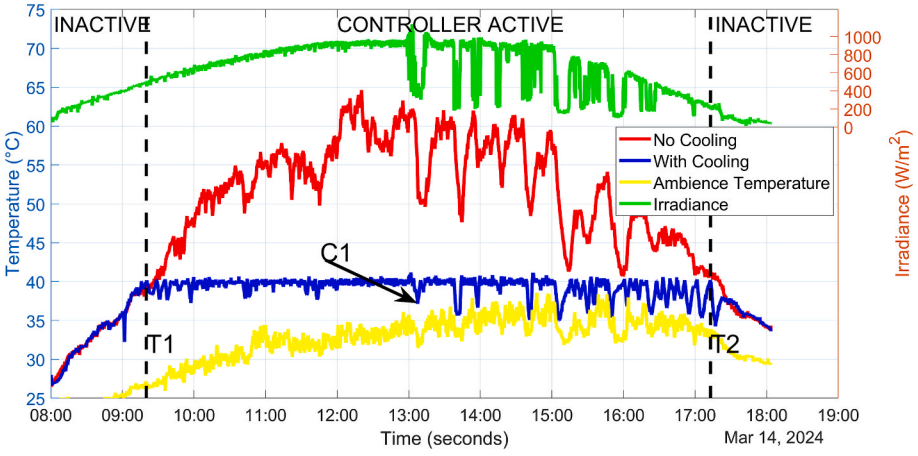


Fig. 14. Temperature comparison of PV modules and irradiance.

the controller is designed to prevent overheating, it does not activate for temperature stabilization below the setpoint, reflecting its dependence on environmental conditions and solar irradiance fluctuations.

The data from March 22, 2024 shown in Fig. 17 demonstrates the temperature regulation of PV modules under mostly sunny irradiance conditions. The uncooled module peaked at 65 °C, while the cooled module maintained a stable 40 °C, effectively preventing overheating. Ambient temperature gradually increased, reaching 33 °C at midday. During the inactive phases before T1 and after T2, cooling was unnecessary. The controller effectively stabilized the PV module temperature throughout the active phase. At Condition C1, a brief irradiance dip caused both modules to experience a temporary temperature drop, highlighting their sensitivity to solar variations. Similarly, during Box 1 (B1), fluctuations in irradiance momentarily lowered the cooled module's temperature below 40 °C, causing temporary deactivation of the cooling system.

The multi-day analysis in Table 7 demonstrates the effectiveness of the SCA-Fuzzy-PID cooling system in regulating PV temperature, improving power output, and optimizing water use under varying weather conditions. The cooled module consistently maintained temperatures around 39.3 °C–39.5 °C, while the uncooled module exceeded 50 °C, peaking at 65.9 °C on the mostly sunny day (Day 3). Despite fluctuations in irradiance, the cooling system enhanced power generation by 9.50 %–10.50 %, with a maximum boost of 13.17 % on Day 1. Voltage stability also improved by 4.56 %–5.18 %, reinforcing the system's role in optimizing PV efficiency. Water consumption varied with thermal demand, ranging from 837.5 L (Day 2) to 1165.8 L (Day 1), demonstrating the controller's adaptability in regulating cooling based on real-time conditions. Overall, the results confirm the cooling system's ability to enhance PV performance while maintaining water efficiency,

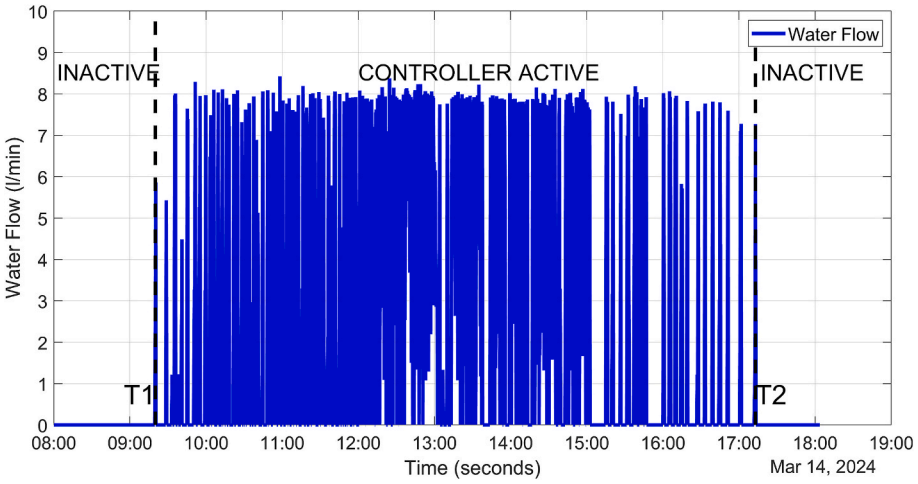


Fig. 15. Water flow rates for controller-based cooling.

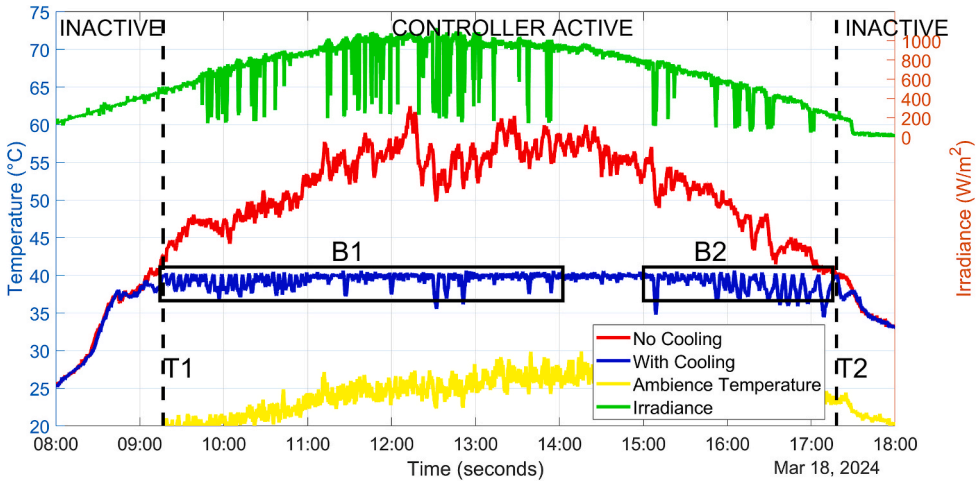


Fig. 16. Temperature comparison of PV modules and irradiance.

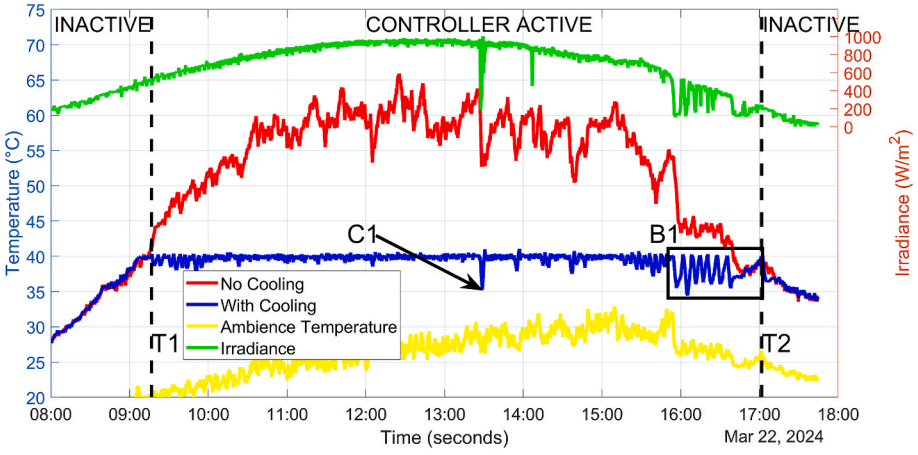


Fig. 17. Temperature comparison of PV modules and irradiance.

making it a practical solution for real-world solar applications.

4. Conclusions

This study presented a self-tuning Fuzzy-PID controller optimized using the Sine Cosine Algorithm (SCA) for active PV cooling. This approach demonstrated superior temperature regulation and power efficiency compared to conventional methods such as free-flow cooling. The experimental results showed that the controller-based cooling system maintained an average PV module temperature of 40.0 °C, leading

Table 7
PV module performance comparative analysis over multiple days.

Parameter		Day 1 (Semi Cloudy)		Day 2 (Cloudy)		Day 3 (Mostly Sunny)	
		No Cooling	With Cooling	No Cooling	With Cooling	No Cooling	With Cooling
Temperature (°C)	Avg	52.3	39.4	51.9	39.3	54.6	39.5
	Max	64.7	41.2	62.5	40.7	65.9	41.1
Irradiance (W/m^2)	Avg	680.3		739.4		737.6	
	Max	1136.7		1131.6		1001.6	
Maximum Output Power (W)	Avg	101.1	110.7	109.4	119.7	108.7	120.0
	Max	174.7	197.7	180.1	198.0	161.6	181.6
Open Circuit Voltage (V)	Avg	32.9	34.4	33.0	34.5	32.8	34.5
	Max	34.37	35.7	34.0	35.9	34.3	35.6
Total Water Consumption (L)		–	1165.8	–	837.5	–	935.6
Total Power Generated (Wh)		1095.11	1199.79	1194.77	1308.16	1151.76	1272.59
Percentage Boost in Power Generated (%)		9.55		9.50		10.50	

to a 12.2 % increase in power output while significantly reducing water consumption by 60 % compared to free-flow cooling. This reduction also implies less pumping energy and resource usage, as the controller dynamically adjusted the water flow based on real-time temperature deviations. In contrast, the free-flow method operates at maximum flow regardless of thermal demand, resulting in unnecessary energy and water waste. A full-day experimental analysis under semi-cloudy, cloudy, and mostly sunny conditions further validated the system's adaptability. The controller successfully adjusted water flow in response to real-time temperature changes, preventing excessive water use while maintaining stable power generation. However, periods of low irradiance resulted in temporary temperature drops below the setpoint, reflecting the system's dependence on environmental conditions. These findings highlight the potential of integrating metaheuristic optimization with fuzzy-PID control for sustainable PV cooling. One practical challenge in real-world implementation is supplying water piping infrastructure efficiently across large-scale solar farms. Future work could explore its implementation on newer PV modules and rooftop installations, as well as hybrid cooling strategies to further improve efficiency.

CRediT authorship contribution statement

Mohamad Shukor Abdul Rahim: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Faridun Naim bin Tajuddin:** Supervision, Funding acquisition, Conceptualization. **Mohd Sazli Saad:** Supervision, Conceptualization. **Md. Hasanuzzaman:** Supervision. **Mohd Zamri Zahir Ahmad:** Visualization, Investigation, Formal analysis. **Ahmad Ashraf Abdul Halim:** Visualization. **E. Che Mid:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank Universiti Malaysia Perlis and the Ministry of Higher Education (MOHE) Malaysia for providing the facilities and financial support Fundamental Research Grant Scheme (FRGS) under a grant number of FRGS/1/2022/TK08/UNIMAP/02/92.

Data availability

The datasets generated during the current study are not publicly available, but are available from the corresponding author at reasonable request.

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